

PUBLIC SUBMISSION

As of: March 21, 2025
Received: March 15, 2025
Status: [REDACTED]
Tracking No. m8a-uej5-ncjf
Comments Due: March 15, 2025
Submission Type: API

Docket: NSF_FRDOC_0001
Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479
Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-3440
Comment on FR Doc # 2025-02305

Submitter Information

Name: Mark McDonald

[REDACTED]

General Comment

See attached file(s)

Attachments

Comment on government AI action plan

Targeted Safeguards to Strengthen National Security and Economic Competitiveness

Dear OSTP,

As a chemical engineer specializing in optimization techniques, I recognize both the transformative potential and the existential risks of artificial intelligence. I urge the Administration to prioritize AI safety regulations to protect American national security and critical infrastructure from **chemical, nuclear, cyber, and biological threats** that could arise from the unchecked development and deployment of advanced AI systems.

The United States must lead in AI innovation while ensuring that bad actors, including both foreign adversaries and domestic threats, do not exploit AI to undermine national security. AI-driven cyberattacks, AI-assisted synthesis of chemical weapons, and AI-enhanced biological threats pose unique risks that demand proactive, strategic intervention. The private sector must remain unburdened by unnecessary restrictions, but targeted, intelligent safeguards will enhance America's AI dominance by preventing catastrophic failures.

To protect America's interests and promote economic competitiveness, I recommend:

1. **AI-Enabled National Security Monitoring** (fighting AI with AI). AI should be leveraged to identify and neutralize threats in cybersecurity, nuclear safety, and biosecurity before they manifest. Investing in AI-driven threat detection will bolster both national defense and private-sector resilience. For example, AI-driven monitoring of wastewater could be used to check for viral DNA suggestive of a biological weapon.
2. **Public-Private Sector Collaboration** on AI Safety Standards. The Administration should foster partnerships with industry leaders and with nonprofit institutes to establish robust AI safety protocols that align with economic and security priorities. Specific institutes that have already dedicated significant resources to this task include the Center for AI Safety (CAIS), the Alignment Research Center (ARC), the Machine Intelligence Research Institute (MIRI), and the Center for Security and Emerging Technology (CSET). Their existing expertise should be leveraged as much as possible to avoid inefficiently duplicating work.
3. **Export Controls on High-Risk AI Applications**. Given the risk of AI-assisted development of chemical and biological weapons, export controls should ensure that U.S.-developed AI models do not fall into the hands of adversaries who seek to exploit them.
4. **Preventing AI-Enabled Biological Threats**. Advanced AI models could dramatically accelerate biological research, including the design of synthetic pathogens. Past incidents, such as journalists acquiring a modified smallpox genome without scrutiny [1], illustrate the underappreciated risks of synthetic pathogen development. With AI advancements, such risks could be magnified. The U.S. should establish targeted oversight mechanisms to prevent the misuse of AI in bioengineering, ensuring that AI-driven research in virology and synthetic biology prioritizes safety and does not facilitate the creation of engineered pandemics.

- a. Of specific worry are “wildfire pandemics,” or pathogens engineered using AI technology to have the *transmissibility of measles and the deadliness of ebola*, which could be used by an adversary to decimate the US population before the medical establishment can mount a response, and “sleeper pandemics,” which have a long enough incubation period that they will spread across the population before we realize they are there.
- 5. Research and Governance for **Advanced Autonomous AI Alignment**. The development of highly advanced AI systems, potentially surpassing human intelligence, poses long-term existential risks. The Administration should invest in research on AI alignment and control mechanisms to ensure that AI remains aligned with human values and national interests, preventing scenarios where AI systems act counter to U.S. security and economic priorities.
 - a. For example, an advanced AI that is given the goal of optimizing financial gains on the stock market might *purposely engineer a catastrophic event* so that it can short the stock market during the resulting downturn. Or, an advanced AI tasked with optimizing military logistics might develop unpredictable escalation strategies. Mandatory safety testing on advanced AI systems, and other guardrails to prevent this misalignment with national interests, are vital to the future success of the country.

By taking a leadership role in AI safety, the U.S. will not only safeguard its people but also strengthen its global standing as the premier hub for AI innovation. A failure to act risks enabling adversaries and criminals to leverage AI against American interests. Smart regulation focused on national security and economic competitiveness is the key to unlocking AI’s full potential while mitigating its most pressing dangers.

Sincerely,

Mark N. McDonald, Ph.D.

- 1. Journalists order smallpox genome from synthetic DNA labs:
<https://www.theguardian.com/science/2006/jun/23/weaponstechnology.guardianweekly>